Anthropogenic Regional Adaptation in Multimodal Vision-Language Model

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Research Questions

- How can vision-language models be adapted to specific regional contexts while maintaining global generalization capabilities?
- What is the optimal trade-off between regional specialization and global performance in multimodal models?
- How effective is model merging combined with regional data filtering for anthropogenic regional adaptation?

Core Idea

Existing vision-language models face a fundamental dichotomy: global models perform well broadly but struggle with underrepresented regions, while regional models excel locally but fail globally. The paper introduces Anthropogenic Regional Adaptation, which systematically balances this trade-off through a Global-Regional Parity (GRP) optimization framework controlled by a globalization factor α derived from real-world globalization indices. The key insight is that rather than choosing between global or regional optimization, models can be adapted through regional data filtering followed by linear model merging, allowing the preservation of global capabilities while gaining regional specificity.

Key Formula

Global-Regional Parity (GRP) Optimization

$$\max_{\theta}[\alpha \cdot Q^{R^{global}} + (1-\alpha) \cdot Q^{R^{regional}}]$$

θ are model parameters, α ∈ [0,1] is globalization factor controlling trade-off, Q^{R^global} measures global performance, Q^{R^regional} measures regional performance. This objective function balances regional adaptation with global generalization preservation.

Linear Model Merging

$$\theta_{merged}(\beta) = \beta \cdot \theta_{regional} + (1-\beta) \cdot \theta_{global}$$

θ_merged are merged model parameters, β controls interpolation weight, θ_regional are region-specific model parameters, θ_global are original global model parameters. This enables combining regional adaptations with global capabilities through weighted parameter interpolation.

Key Figure

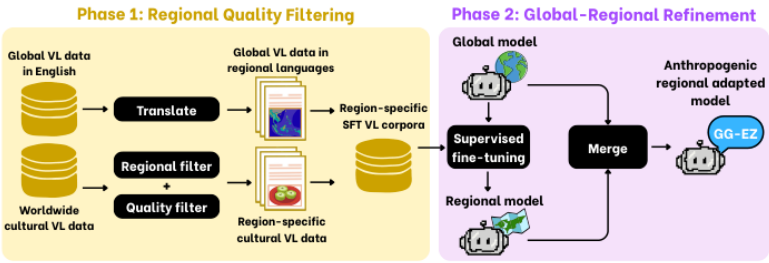

Fig. 2: Overview of our Geographical-generalization-made-easy (GG-EZ) framework. Our framework consist of 3 constituents: (1) High-quality regional data filtering pipeline; (2) supervised fine-tuning to create a high quality regional-specific model; and (3) model merging to capture the best combination between regional-specific and global represention while also maintaining the generalization capabilities of the model.

objective is constructed as a weighted combination:

$$\max_{\theta} \left[\alpha \cdot Q^{\mathcal{R}^{\text{global}}} + (1-\alpha) \cdot Q^{\mathcal{R}^{\text{regional}}} \right]$$

This formulation creates a single scalar objective that can be optimized using standard techniques, with α serves as a critical control mechanism that can be adjusted based on the specific focus and characteristics of each region. For regions with strong global connections or dependencies, α can be set closer to 1 to emphasize maintenance of $\mathcal{R}^{\text{global}}$ conditions, while for regions where local impacts dominate, α can approach 0 to prioritize $\mathcal{R}^{\text{regional}}$ optimization. The solution achieves "best parity" between regions, representing the most favorable equilibrium where neither aspect is disproportionately favored according to our specified weighting scheme.

4 Geographical Generalization Made Easy (GG-EZ)

Geographical-generalization-made-easy (GG-EZ) is a general framework for regional adaptation of global multimodal models that achieves effective region-specific performance while preserving original generalization capabilities. As illustrated in Figure 2, the core mechanism of GG-EZ consists of two phases: **regional quality filtering** and **global-regional refinement**. GG-EZ not only enables adaptation across local textual and imagery contexts while maintaining base model capabilities, but also better yet being architecture-agnostic, allowing effective regional adaptation across different vision-language architectures.

Figure 2: Illustrates the complete GG-EZ framework showing the two-phase approach of regional quality filtering followed by global-regional refinement through model merging.

Limitations

- Method requires manual threshold tuning for regional data filtering and reward model scoring
- Globalization factor α derivation depends on external KOF index which may not capture all cultural nuances
- Linear model merging may not be optimal for all architecture types and could lose some specialized regional knowledge
- Evaluation limited to Southeast Asian region - generalizability to other underrepresented regions unclear